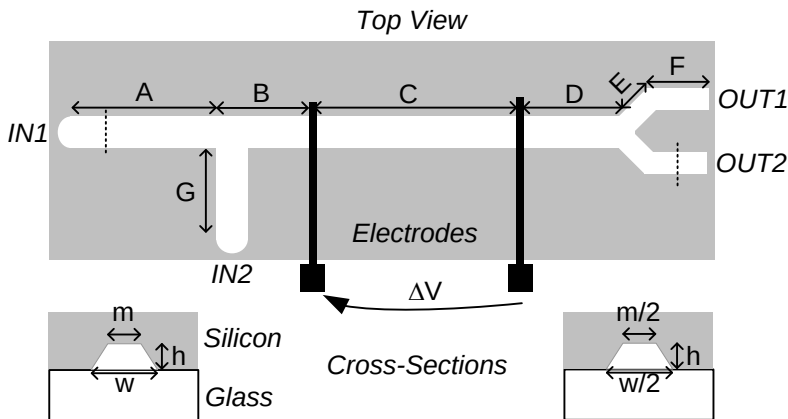


Politecnico di Milano
Biochip A.A. 2013/2014 - M. Carminati
Appello 22.09.2014

*Test duration: 2 hours - No support material (textbooks, laptops, smartphones, notes) admitted.
 All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.*

Exercise 1

Consider the following microfluidic system (with 2 inputs and 2 outputs) realized by etching a channel with trapezoidal cross-section in Silicon. Sections A-B-G represent a T-junction, section C is between two electrodes fabricated on the glass substrate, D-E-F represent the bifurcated output section. Adopt proper approximations.

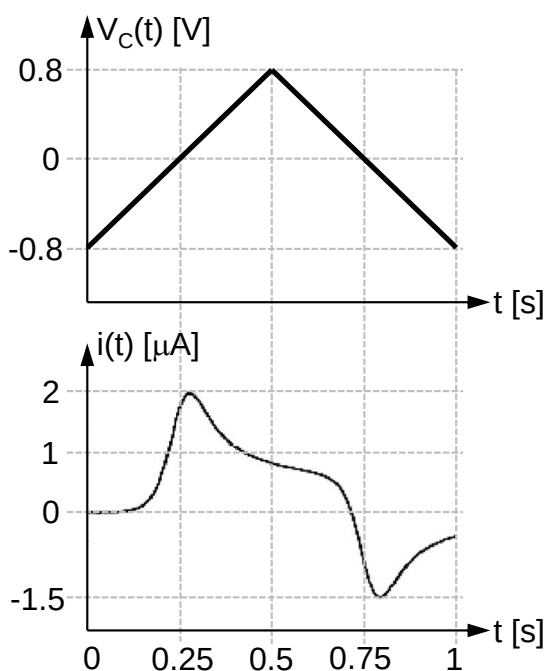


$A = 1\text{mm}$
 $B = 0.5\text{mm}$
 $C = 1\text{mm}$
 $D = 0.2\text{mm}$
 $E = 0.2\text{mm}$
 $F = 0.5\text{mm}$
 $G = 0.2\text{mm}$
 $w = 50\mu\text{m}$
 $m = 30\mu\text{m}$
 $h = 30\mu\text{m}$
 $\eta_{\text{water}} = 10^{-3}\text{ Pa}\cdot\text{s}$
 $\xi = 50\text{mV}$ (zeta potential)
 $P_0 = \text{atmospheric pressure} = 101\text{kPa} = 1\text{atm}$

- If $P_{\text{IN}1} = 109\text{kPa}$, $P_{\text{OUT}1} = P_{\text{OUT}2} = P_0$, find $P_{\text{IN}2}$ in order to achieve a flow velocity of $1\text{mm}/\text{min}$ in section C. Determine also the flow velocities in all the sections.
- How does the velocity in section C changes if output 2 gets clogged?
- If water enters into IN1 and PBS enters into IN2, find and draw the liquid separation (cross-section) in section C.
- Considering a PBS solution filling the channels, find the voltage difference ΔV to be applied to the electrodes in order to obtain in section C the same fluid velocity of point (a) by means of *electro-osmotic* flow.
- Now consider a spherical particle of diameter $10\mu\text{m}$ and charge $Q = 1000$ electrons in deionized water, find the particle direction and velocity if $\Delta V = +10\text{V}$.

Exercise 2

Considering the following waveforms corresponding to one cycle of a *cyclic voltammetry* (CV) three-electrode measurement with a FeCN redox species in concentration $10\mu\text{M}$: $V_C(t)$ is the voltage applied to the electrochemical cell (with respect to Ag/AgCl RE) and $i(t)$ is the current sensed at the working electrode (WE):



- Determine the *scan-rate* of this CV.
- Plot the *voltammogram* (with quantitative axes).
- Determine the values of the reduction (E_1) and oxidation (E_2) potentials (with respect to the Ag/AgCl reference electrode). Is the reaction reversible?
- Given this *scan rate*, determine the suitable bandwidth of the potentiostat.
- Considering a max. tolerable error of 10% of $\Delta E = E_1 - E_2$ in ideally reversible conditions, what is the accuracy required for the voltage generator source (V_C)?
- Draw the scheme of the current reader connected to the WE and choose the value of the feedback resistor of the transimpedance amplifiers considering a dual supply voltage of $\pm 5\text{V}$.
- If the concentration of the FeCN redox species is reduced to $1\mu\text{M}$, how does the voltammogram change?
- If a small-signal impedance measurement is performed on this electrochemical system, find the DC bias potential V_C at which the lowest charge transfer resistance (R_{CT} in this Faradaic interface) is measured.
- If the electrodes for this CV measurement are to be inserted into a miniaturized biochip and thus fabricated in a standard CMOS microelectronic process, what are the issues from the materials point of view.